

ZILING ZHANG

Data Scientist / Edge AI Robotacist / C++ Python expert

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zilingz

US Permanent Resident

EXPERIENCE

Advisory Data Scientist

Automation Technical Implementation Group | Ricoh Graphic Communications

Aug 2022 – present

Boulder, CO

- (Ongoing) Build printshop digital twin with CAD and Luma AI NeRF usdz models in Nvidia Omniverse, for training field engineers and Isaac Sim warehouse robotics.
- Advise on printer data producers design, enabling stateless event streaming.
- Develop serverless ETL pipeline aggregating printer/sensor timeseries data, prep for cloud training, inference, visualization. AWS Athena, lambda, SDK for pandas.
- Perform printhead predictive maintenance inference with SageMaker serverless.
- Architect AI model edge deployment for air-gapped customers, Nvidia Jetson (aarch64).

Staff/Advisory Imaging Science Engineer

Boulder Development Section | Ricoh Graphic Communications

Aug 2019 – Aug 2022

Boulder, CO

Computer Vision + Automation R&D for Continuous Feed Industrial Printer

- Detect nozzle defect by training a semantic segmentation Deeplab v3+ neural network, accelerate with Nvidia TensorRT 8 int8 quantization, achieve 37x speedup.
- Prototype nozzle uniformity algorithms, accelerate with SIMD and threading, achieve 40x speedup. Intel MKL/TBB, OpenCV, Visual Profiler, Concurrency Visualizer.
- Architect image processing pipeline that can withstand 7GB/s scanner data, perform realtime defect detection and compensation to achieve superior print quality.
- Standardize the team's coding practice, enforce modern C++ RAII style, C++ Core Guidelines, Google python style / pylint, modularize code into testable liblets.
- Setup CMake, google test, GitLab CI, docker runners for high velocity iteration.
- Filed 28 patents, granted 10, achieved Master Inventor title, fastest in Ricoh Boulder.

Robotics Research Assistant

Human Centered Robotics Lab | Colorado School of Mines

Aug 2018 – Dec 2019

Golden, CO

- DARPA subteranean challenge Mines team, SLAM LiDAR 3D mapping ROS node.
- Toyota InfoTechnology Center, V2X IoT multi-vehicle collaborative perception.
- Metcalf Archaeology, long term place recognition AR iOS app (Swift, OpenCV C++).
- Kinect + FLIR lpton sensor fusion, forehead temperature measurement TurtleBot.

Scientific Translator (English -> Chinese)

Contract based

Jan 2012 – Aug 2018

Remote

- (NetEase Inc.) Translate OpenCourseWare, Coursera lectures subtitles, 1.87 million words.
- (Yeeyan.org) Book translation, Net Smart by Howard Rheingold, MIT press.

Software Developer

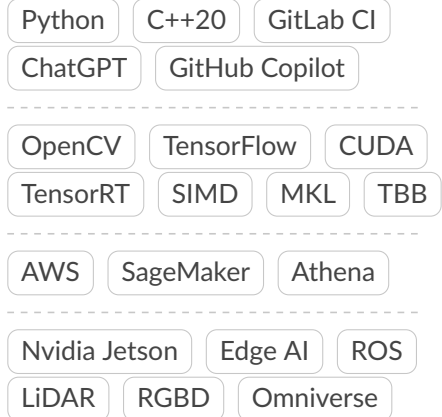
JindianTuteng Talent information solution Inc.

Oct 2011 – June 2012

Guangzhou, China

- Rule based transaction fraud detection for Foreign Exchange Management. Java, J2EE, JSP, IBM WebSphere, IBM DB2.

SKILLS



EDUCATION

M.S. Computer Science

Colorado School of Mines

Aug 2018 – Dec 2019

Robotics, HPC ★ GPA 3.96

Courses: Computer Vision, Parallel Computing, Robotics, Computer Architecture

Paper: Collaborative Localization for Occluded Objects in Connected Vehicular Platform
4th author, 2019 IEEE 90th Vehicular Technology Conference

B.S. Applied Physics

University of Science and Technology of China

Sep 2007 – June 2011

Condensed Matter Physics

Paper: Speed of Last Vehicle Feedback Strategy in Intelligent Transportation Systems
3rd author, International Journal of Modern Physics C, 2011

INTERESTS

